Zeynep Tandogan

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EDUCATION

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE (EPFL), Lausanne, Switzerland **Sept 23 - Present**
MSc. in Computer Science

- Expected graduation date: June 2025
- Specialization in Data Analytics

SABANCI UNIVERSITY, Istanbul, Turkey
B.S. in Computer Science and Engineering, Faculty of Engineering and Natural Sciences **Sept 17 – June 22**
B.S. in Electronics Engineering, Faculty of Engineering and Natural Sciences – Double Major **Sept 17 – June 23**

- Double Major in Electronics Engineering & Minor in Business Analytics
- Distinctive student with comprehensive 100% scholarship, CGPA: 3.96/4
- Learning assistantships in Advanced Programming (CS204) and Natural Sciences 202 (NS202) which involved conducting homework assistance and lesson clarification for CS204, and guiding discussions, assessments, and skill development in NS202.

WORK EXPERIENCE

TELESCOPE LABS
Machine Learning Engineer, London, United Kingdom **Oct 22 - Sept 23**

- Collaborated on the deployment of Vantage, a chatbot innovation by Telescope Labs
- Integrated interpretable probabilistic modeling alongside extensive language models to enable comprehensive explanations of virtual economies
- Strengthened functionality through the incorporation of semantic search algorithms
- Enriched the chatbot's skillsets, including the addition of features like a chart builder and data reasoning capabilities
- Used Python and JavaScript as languages and performed tasks by using Azure tools such as Azure Data Explorer (with KQL).

GETİR
Data Science Intern, Istanbul, Turkey **March 22-Nov 22**

- Implemented SQL queries and Python for data analysis
- Developed Machine Learning models to increase user satisfaction by predicting missing items in user's shopping baskets
- Implemented arrangements to solve business problems and created reports for business units
- Worked on API services
- Used Python and SQL as languages and performed tasks by using Redshift, Git, Bitbucket, Airflow, JupyterHub and CloudWatch tools.

HUAWEI TURKEY R&D CENTER
Artificial Intelligence Intern, Istanbul, Turkey **Aug 21 – Sept 21**

- Applied pretrained Named Entity Recognition (NER) models to Arabic queries as a part of Petal Search project
- Arranged Arabic NER datasets with pre-processing and post-processing techniques and prepared mixed datasets with English entities
- Used Python as a programming language; Pycharm, Colab and AWS as development environments

ASELSAN
Software Engineering Intern, Ankara, Turkey **June 21 – Aug 21**

- Developed a website that that allows multiple users to watch real-time broadcasts from the cameras they have chosen to watch simultaneously
- Used React for Front-end implementation and Spring Boot for Back-end implementation
- Worked with WebRTC and Kurento for real-time communication

GARANTI BBVA TEKNOLOJI
Intern Enterprise Communications Network Engineer, Istanbul, Turkey **Aug 20 – Sept 20**

- Applied Robotic process automation (RPA) and Selenium to automate regular, time-consuming activities
- Used Cisco ICM advanced scripting for communication
- Worked on SQL programming in data analysis

PROJECTS

Reinforcement Learning for Language Model Alignment - SwissAI (EPFL) **Feb 24- Present**

- Worked as a student researcher that is supervised by Prof. Caglar Gulcehre, under the scope of SwissAI project in CLAIRE Lab.
- Aimed for simplicity and efficiency in aligning large language models with human preferences (RLHF).

Transcription of Ottoman Machine-Print Documents **Jan 21-July 23**

Project supported by The Scientific and Technological Research Institution of Turkey

- Developed machine learning model for transcription from Ottoman to Turkish
- Performed segmentation of the pages by using computer vision techniques
- Implemented a GUI for ease of transcription
- Language & Libraries & Tools: Python, OpenCV, Pytorch, Pandas

User Interaction Mechanism for Multimodal Interfaces in Collaborative Analytical Sessions **Sept 21 – June 22**

Bachelor Graduation Project

- Performed real-time gesture detection and emotion detection with the stream taken with Kinect.
- Implemented speech-to-text translation into the conversation of participants of live session.
- Language & Libraries & Tools: Python, OpenCV, Pykinect2, Pygame, Norfair and Kinect
- Github link: https://github.com/zeyneptandogan/HandGesture_Kinect

System-Theoretic Analysis of Nano-Communication Signals **Sept 20 – Feb 21**

Voluntary Project

- Researched academic articles to get detailed information about molecular communication, obtained prediction result by using machine learning (linear regression and neural network)
- Language & Libraries & Machine Learning Algorithms: MATLAB, Python, NumPy, Pandas, matplotlib, Linear regression, Neural Network

PUBLICATIONS

- Bilgin Tasdemir, E. F., Kızılırmak, F., Akcan, M. A., Kuru, M., **Tandogan, Z**., Akansu, S. D., & Yanıkoglu, B. (2022). Transcription of Ottoman Machine-Print Documents. Research Square. https://doi.org/10.21203/rs.3.rs-2273629/v1

SKILLS AND ABILITIES

Programming Languages: Python, C++, JavaScript , MATLAB
Tools: Git, DataGrip, JupyterHub, MySQL, Azure Data Explorer, Bitbucket, Airflow, AWS Redshift
Languages: Turkish (Native), English (Advanced), French (Beginner)
Other technical skills: React, Spring Boot, WebRTC, Docker, Flutter, Robotic Automation Process (RPA)

ACHIEVEMENTS

- ALES score of 98/100 (Graduate education entrance exam (equivalent to GRE in Turkey), placing 28th among all participants over 120.000 people
- IELTS Score: 7.5 Overall
- Comprehensive 100% scholarship in the university education for success in the University Entrance Exam and 100% tuition waiver in the high school education for success in High School Entrance Exam